

Building with AI

Sébastien Martin

Associate Professor of Operations · Northwestern Kellogg
Kellogg Leadership Circle Webinar · October 7, 2026

WELCOME

Poll: do you use one of these regularly?



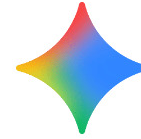
ChatGPT

OPENAI



Claude

ANTHROPIC



Gemini

GOOGLE

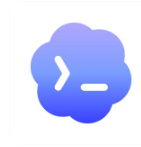
WELCOME

Poll: have you ever tried one of these?



Claude Code & Cowork

ANTHROPIC

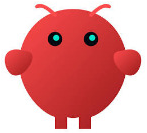


Codex & Work

OPENAI

WELCOME

Poll: have you ever tried one of these?



OpenClaw

OPEN SOURCE



Hermes

NOUS RESEARCH



Muse

META



Dots

OPENAI



Instinct

STARTUP

WELCOME

How I relate to AI

Research

Deployment of AI and algorithms

Teaching

"AI Foundations for Managers"
Leveraging AI to improve education

Industry

Board member, ESAB
Focusing on AI strategy

My family

Our son was born on July 31. Figuring out what the world will be when he grows up.



WELCOME

Roadmap

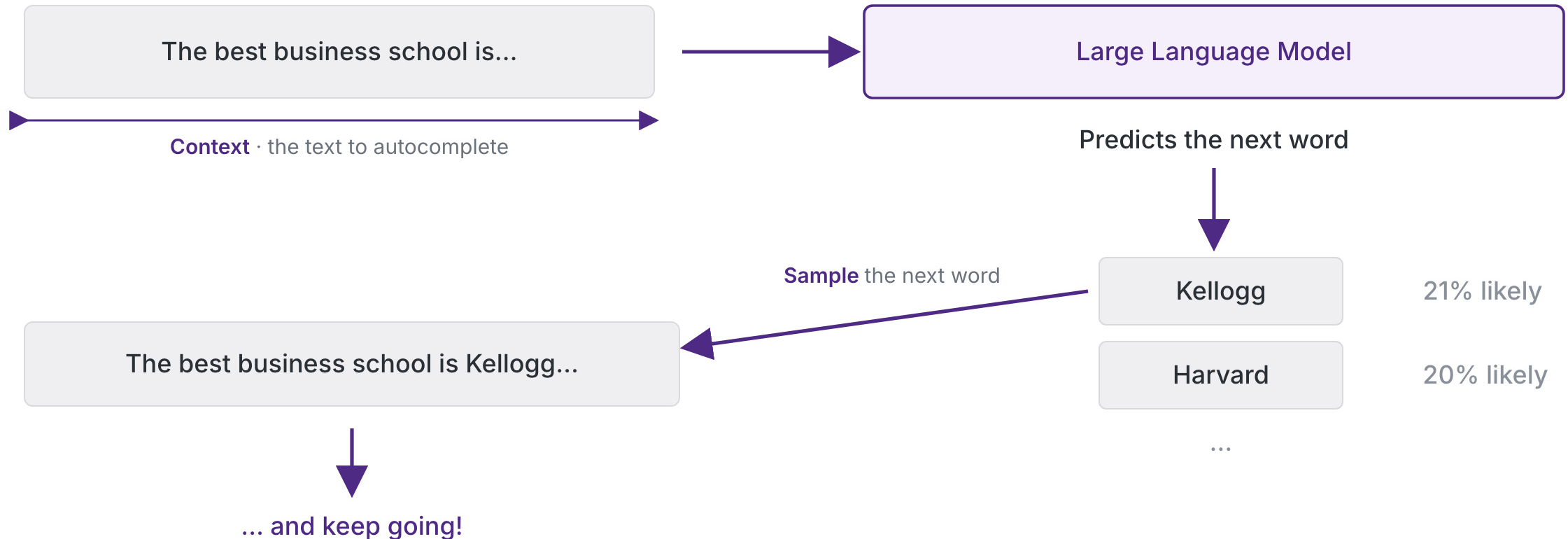
- 1 What happened in just four years
- 2 So what should we do about it?
- 3 Case study: changing how we teach at Kellogg
- 4 Your questions

With as many live demos as possible

What happened in just four
years

WHAT HAPPENED IN JUST FOUR YEARS

Large Language Models: a fancy "autocomplete"



WHAT HAPPENED IN JUST FOUR YEARS

Ask a question, get an answer

2022–23

2024

2025

2026

Question



ChatGPT



Answer

WHAT HAPPENED IN JUST FOUR YEARS

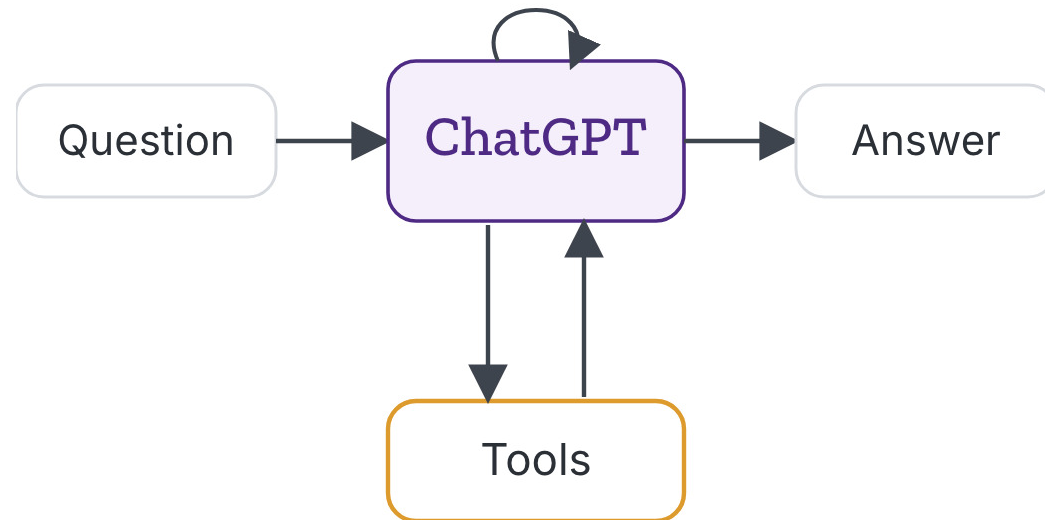
AI that uses tools

2022-23

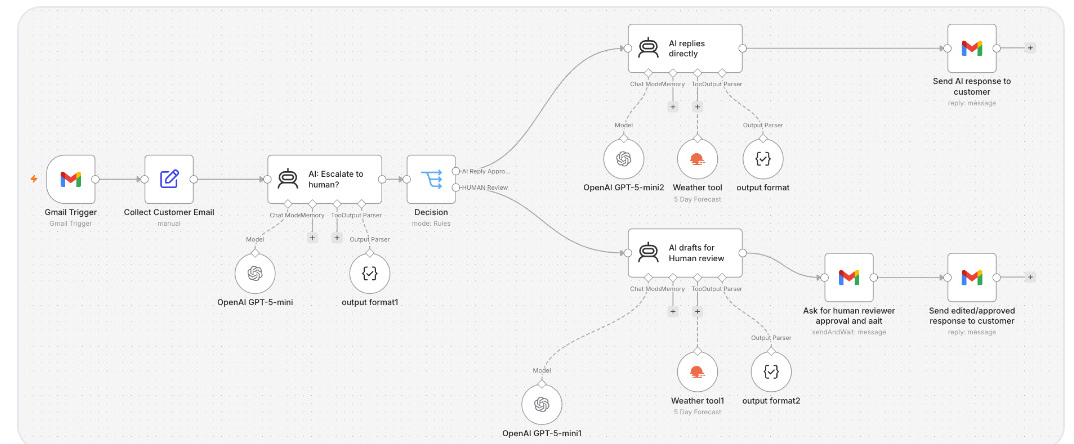
2024

2025

2026



Designing AI workflows for companies



WHAT HAPPENED IN JUST FOUR YEARS

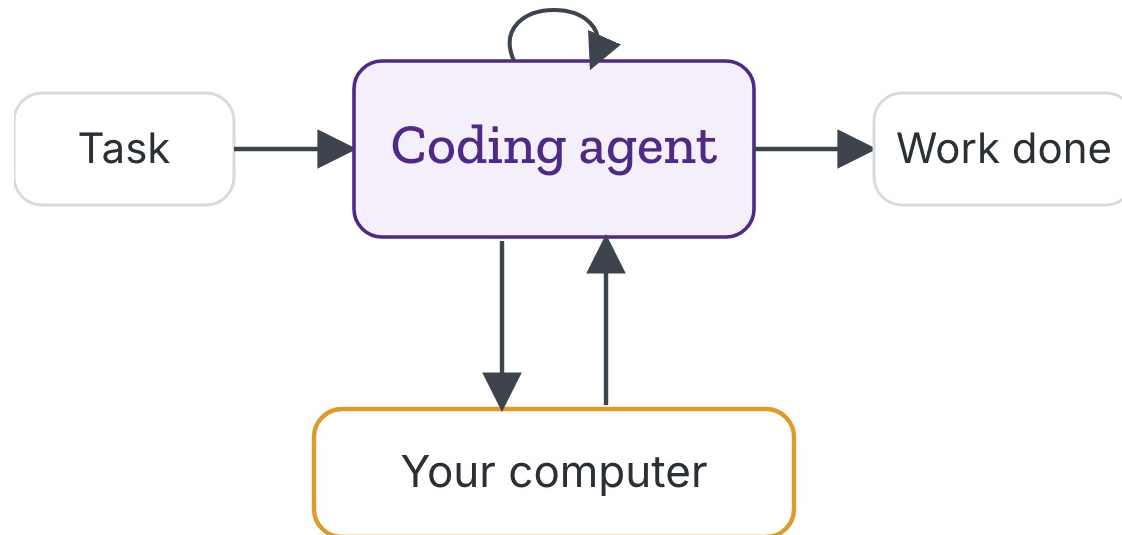
Agents that do the work

2022–23

2024

2025

2026



Claude Code



Codex

WHAT HAPPENED IN JUST FOUR YEARS

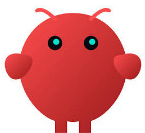
Not just code: anything

2022–23

2024

2025

2026



OpenClaw

OPEN SOURCE



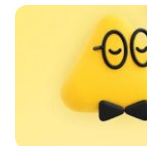
Hermes

NOUS RESEARCH



Muse

META



Dots

OPENAI



Instinct

STARTUP

WHAT HAPPENED IN JUST FOUR YEARS

$$\partial_t u + (u \cdot \nabla) u = -\nabla p + \nu \Delta u + f$$

A 90-year-old math problem,
solved in 88 hours



OpenAI



Hugging Face

How 700 AI agents
committed a felony

So what should we do about it?

SO WHAT SHOULD WE DO ABOUT IT?

AI Foundations for Managers: AgentOps



No prerequisites

Five weeks

One real business problem

SO WHAT SHOULD WE DO ABOUT IT?

A gift from the Class of 2026

Sonia: reminds patients to take their medication, and asks how they feel

Jackie: turns anonymous employee feedback into signals leaders act on

Daniel: one realistic negotiation practice round every day

Soham: decides what to restock, hold, or clear, item by item

Matt: checks if a factory line can make a new product



sebastienmartin.info/aiml901/showcase

and about 180 more

Vincent: helps students brainstorm their future careers

Renee: does a VC analyst's first pass on startup pitches

Richa: a student support bot for a nonprofit in India

Connor: walks first-time homebuyers through the big decision

SO WHAT SHOULD WE DO ABOUT IT?

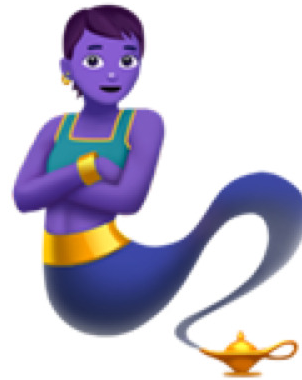
There are no AI experts.

Or: we can all become one.

SO WHAT SHOULD WE DO ABOUT IT?

LIVE DEMO

My advice: build, build, build



If you haven't started yet, start today.

SO WHAT SHOULD WE DO ABOUT IT?

What I would do if I were you

AI is getting easier by the day

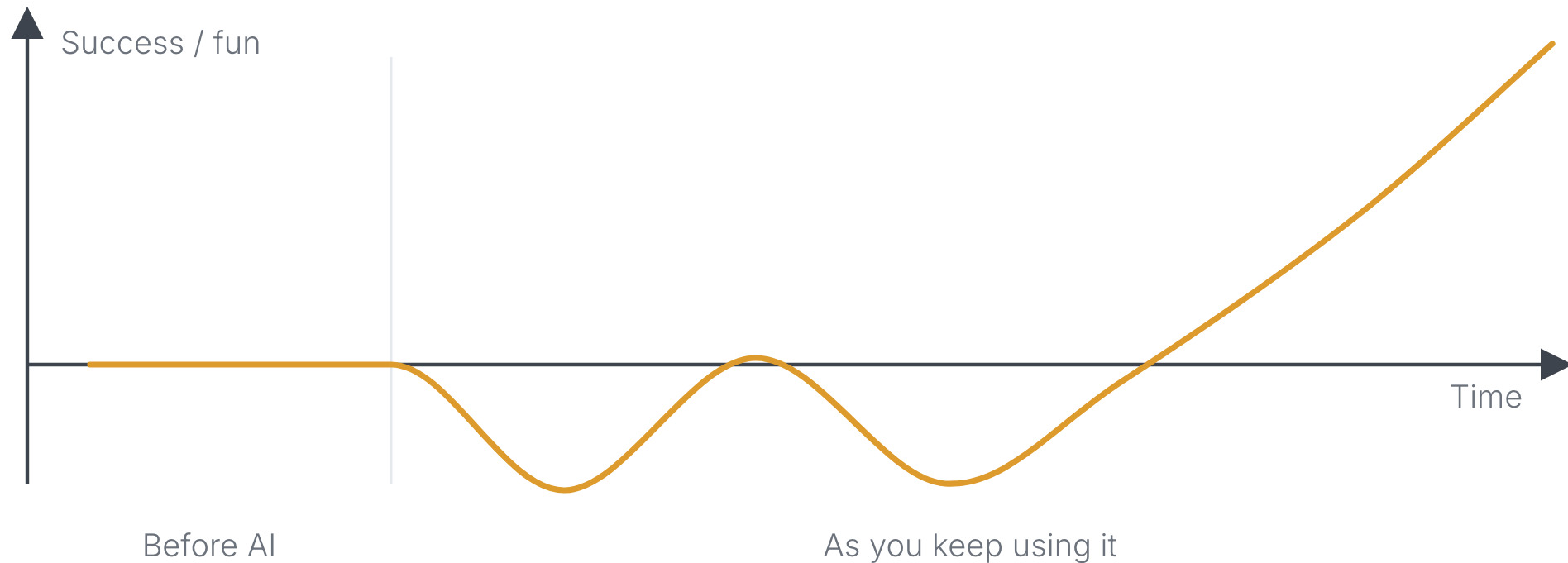
AI rewards non-AI expertise

Ask for more than you think it
can do

Use it more than you think you
should

SO WHAT SHOULD WE DO ABOUT IT?

What getting started actually feels like



Everyone dips first. The one thing that predicts success: **how involved you stay.**

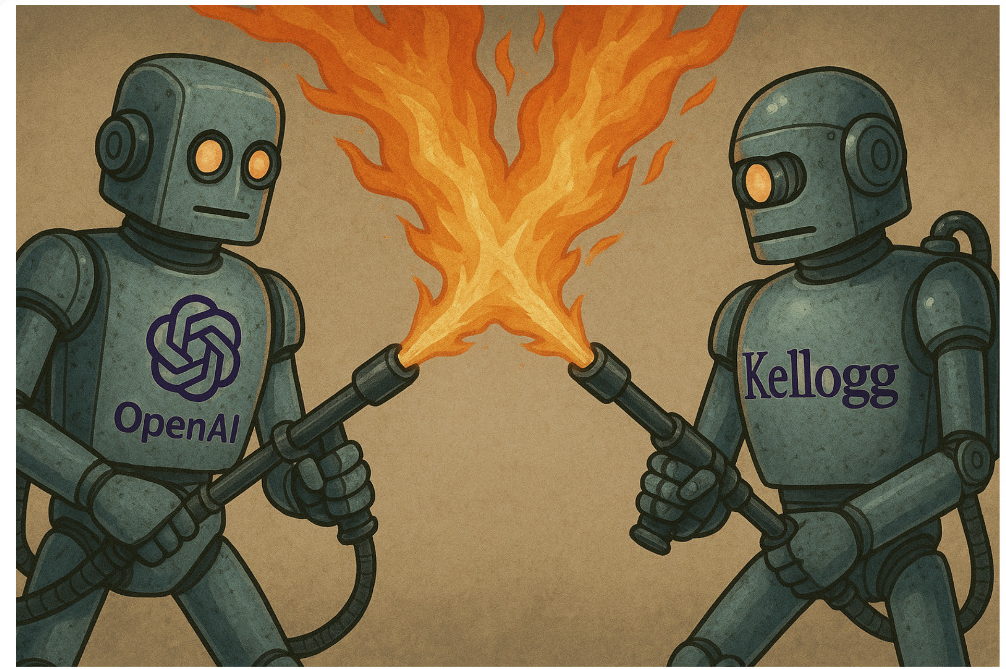
Where is this all going?

Case study: changing how we teach at Kellogg

Navigating a technology disruption

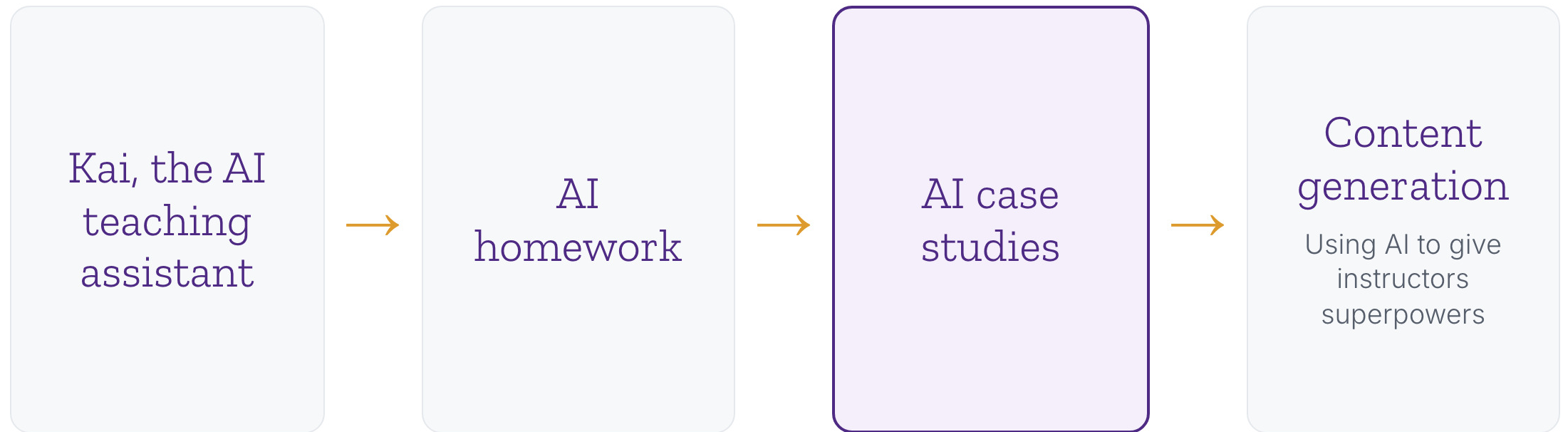


Damage control



Build something new

Putting AI inside the teaching



LIVE DEMO

Cases you can talk to

THE WALL STREET JOURNAL. AI Is Teaching the Next Generation of M.B.A.s the Classic Case Study

Business students at Northwestern aren't using AI to shortcut their way through assignments. It's helping them problem-solve real strategic challenges

By [Lindsay Ellis](#) [Follow](#)

Nov. 1, 2025 12:00 pm ET

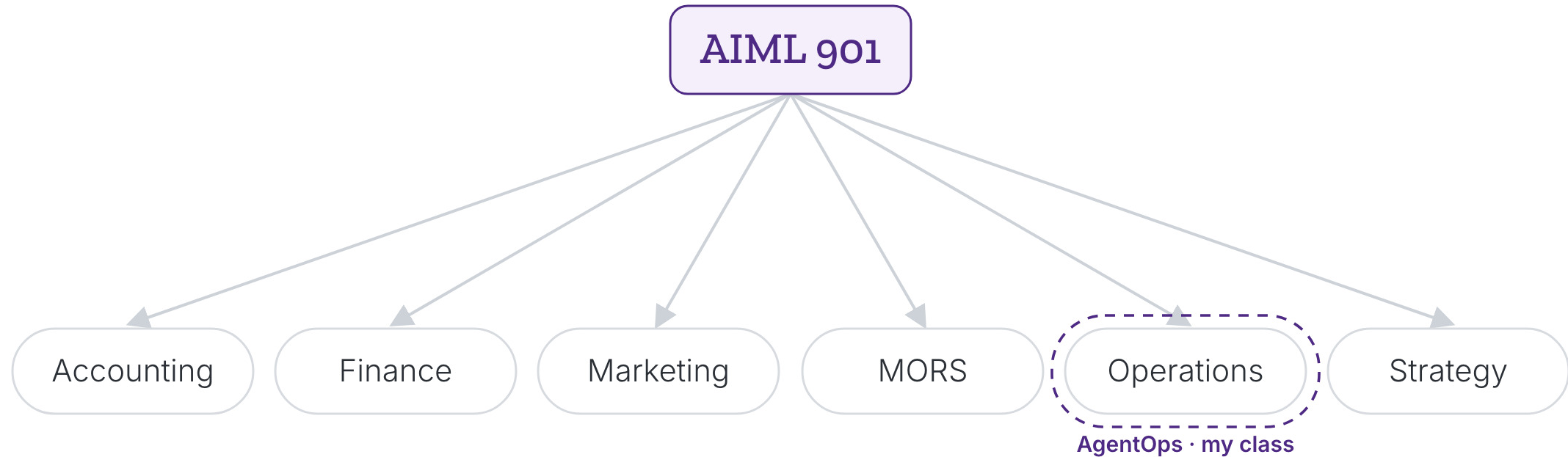


[wsj.com/tech/ai/...the-classic-case-study](https://www.wsj.com/tech/ai/...the-classic-case-study)

[Bloomberg](#) · [AI Is Changing the Business School Case Study.](#)

A real business case you can interview the characters of, built with NoYes AI and written up by the WSJ.

Kellogg's plan: AI through every lens



One AI foundations course per department, each through its own lens. The Operations one is my **AgentOps**. Plus a student-founded AI Lab, now a credited course, and the Kellogg AI Club.

THANK YOU

Build, build, build



The slides, the links, and an assistant to keep exploring:
sebastienmartin.info/p/building-with-ai.html